

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 3.2: Neural Networks — Architecture and Activations

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Topic 3.2 Agenda

1. **Neural Network Architecture** — neurons, layers, depth
2. **Activation Functions** — sigmoid, tanh, ReLU, universal approximation

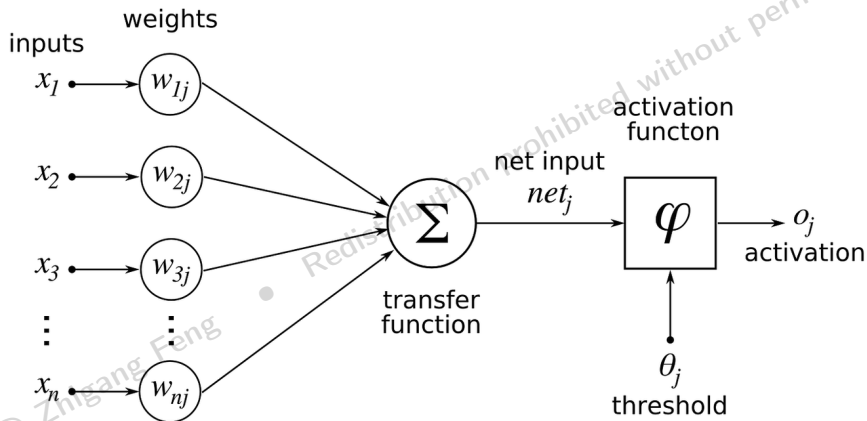
Arc: T1 Why ML → T2 NN Concepts → T3 Training + Backprop → T4 Specialized + Lab

Neural Networks: Architecture and Activation Functions

Neural Networks: The Basic Building Block

- Artificial neural networks are:
 - Inspired by biological neural networks
 - Composed of interconnected neurons organized in layers
 - Trained by adjusting weights based on data
- **Key components:**
 - **Neuron:** Receives inputs, applies an activation function, produces output
 - **Activation function:** Nonlinear map applied by each neuron (sigmoid, tanh, ReLU)
 - **Loss function:** Measures how well predictions match true values — the objective to minimize

Single Neuron: Visual



Single Hidden Layer Network

- Input $x \in \mathbb{R}^M$, output $y \in \mathbb{R}^N$.
- The j -th hidden unit:

$$h_j^{(1)} = f_j^{(1)}(x) = \sigma \left(b_j^{(1)} + \sum_{m=1}^M w_{j,m}^{(1)} x_m \right), \quad j = 1, \dots, U_1 \quad (1)$$

- The i -th output:

$$y_i = \sum_{j=1}^{U_1} \tilde{w}_{i,j} h_j^{(1)} + \tilde{b}_i, \quad i = 1, \dots, N \quad (2)$$

- σ is a nonlinear activation function.
- Common choices: $\sigma(x) = \tanh(x)$, $\sigma(x) = \max(0, x)$ (ReLU), $\sigma(x) = \frac{1}{1+e^{-x}}$ (sigmoid).

Two Hidden Layer Network

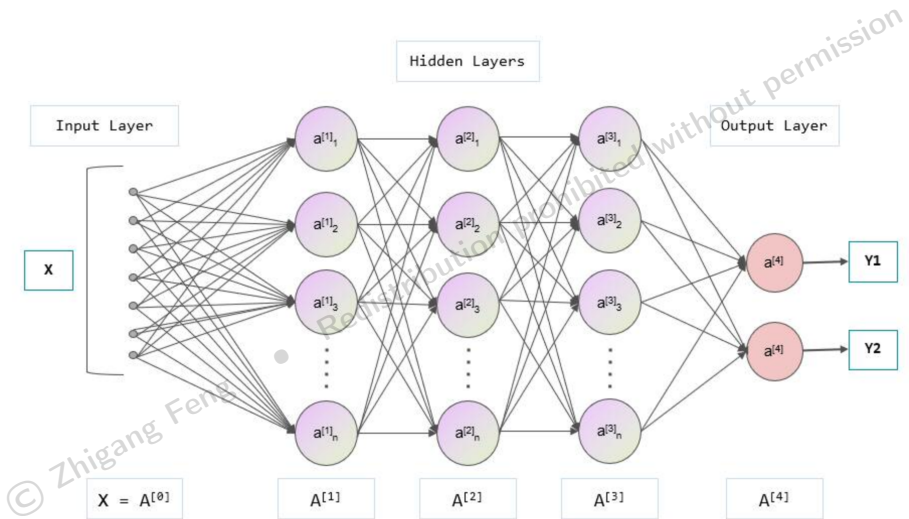
- Take the linear combination of $h^{(1)}$ and pass through another activation:

$$h_j^{(2)} = \sigma \left(b_j^{(2)} + \sum_{k=1}^{U_1} w_{j,k}^{(2)} f_k^{(1)}(x) \right), \quad j=1, \dots, U_2 \quad (3)$$

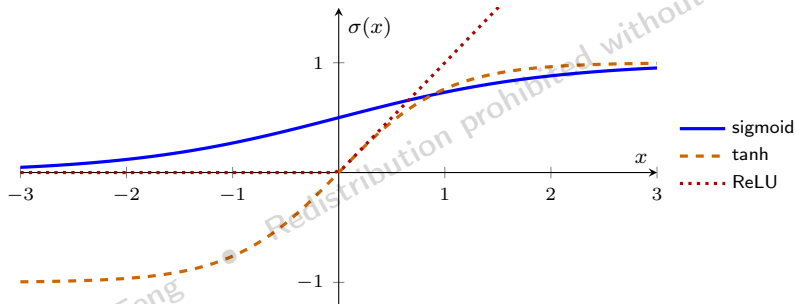
$$y_i = \sum_{j=1}^{U_2} \tilde{w}_{i,j} h_j^{(2)} + \tilde{b}_i, \quad i=1, \dots, N \quad (4)$$

- Pattern:** Each layer performs a linear transformation followed by a nonlinear activation. Depth adds representational power.
- Macro analogy:** Successive layers are like successive time steps in a dynamic program — each transforms the “state” into a richer representation.

Hidden Layers: Visual



Activation Functions: Visual



Sigmoid & tanh saturate ($\Rightarrow$ vanishing gradients); ReLU is piecewise linear ($\Rightarrow$ sparse activation, fast gradient flow).

Activation Functions: What They Do

- **Sigmoid:** $\sigma(x) = \frac{1}{1+e^{-x}}$ — outputs in $(0, 1)$, useful for probabilities.
- **Tanh:** $\tanh(x)$ — outputs in $(-1, 1)$, zero-centered, smoother than ReLU.
- **ReLU:** $\max(0, x)$ — simple, sparse activation, dominant in modern networks.
- **Why nonlinearity matters:** without activation functions, stacked layers collapse into one linear map — depth gains us nothing without nonlinearity.
- **For economists:** the activation is the network's analogue of choosing a flexible functional form (translog, CES, semi-parametric) — but the network composes it across layers.

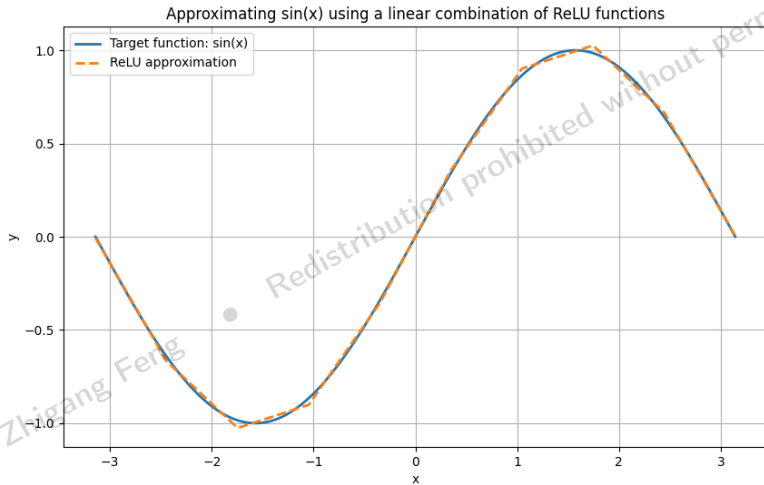
Function Approximation with ReLU

- **Goal:** Approximate arbitrary nonlinear functions using simple building blocks.
- **Key Idea:** Linear combination of shifted ReLU functions:

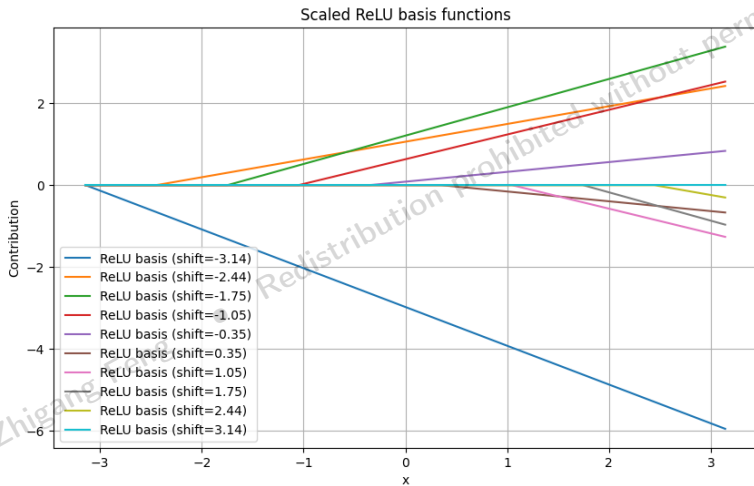
$$\hat{y}(x) = a_0 + \sum_{i=1}^N a_i \text{ReLU}(x - b_i)$$

- **Shifts** b_i : Define the “kink” points where marginal slope changes.
- **Coefficients** a_i : Scale each ReLU’s contribution to match local slope.
- **Example:** Approximating $\sin(x)$ on $[-\pi, \pi]$:
 - Choose shifts b_i evenly over the domain
 - Form design matrix with columns $\text{ReLU}(x - b_i)$ and a bias
 - Solve a least-squares problem for coefficients a_i
- **Economist’s take:** ReLU kinks are piecewise-linear splines — the network *learns* the knot locations, unlike a fixed Chebyshev grid.

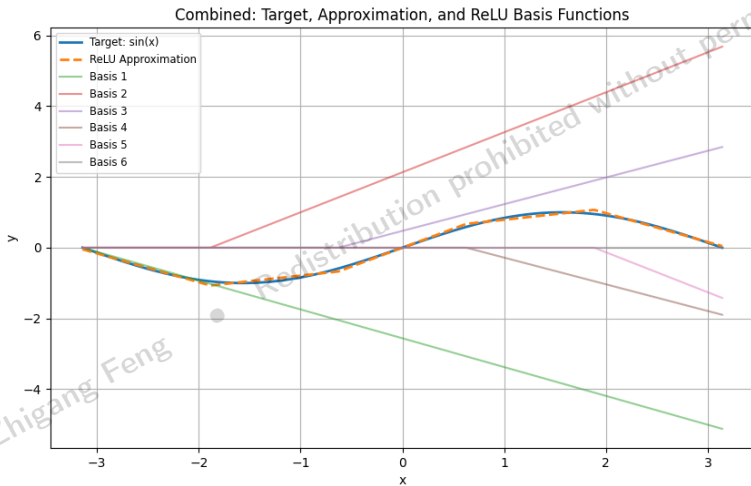
ReLU Approximation: Increasing Complexity



ReLU Approximation: More Kink Points



ReLU Approximation: High Fidelity



Why Neural Networks Work: Universal Approximation

Universal Approximation Theorem (Cybenko 1989; Hornik 1991)

Any continuous function on $[0, 1]^d$ can be approximated *uniformly* by a two-layer network — for sigmoidal σ (Cybenko), for any bounded nonconstant σ (Hornik), indeed for *any* non-polynomial σ (Leshno, Lin, Pinkus & Schocken 1993).

Intuition — you just watched the proof. Those ReLU plots *were* the construction: a weighted sum of shifted kinks $\sum_i a_i \text{ReLU}(x - b_i)$ bends to fit any curve, and a sigmoid network stacks soft “bumps” the same way. **One hidden unit = one building block; add units $\Rightarrow$ shrink the error.** Nonlinearity is the glue — drop it and the whole sum collapses back to a single straight line.

Existence, not a recipe. UAT promises a good network *exists* — not that training will *find* it, nor that “enough” units is a *small* number. Approximation $\neq$ optimization $\neq$ generalization; T3 takes on the finding. (*The same theorem later is why transformer attention approximates — Lec. 7.*)

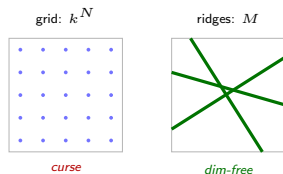
Why Neural Networks Work: Beating the Curse of Dimensionality

Barron (1993): a dimension-free rate

A one-hidden-layer network reaches squared error of order $O(1/M)$ in M units — *independent of the input dimension N* . Fixed series bases (polynomials, splines, Chebyshev) manage only $O(1/M^{2/N})$: to hold accuracy as N grows, M must blow up exponentially.

Why the gap?

- **Grid basis is fixed:** resolution k per axis needs k^N pieces — most parked where f barely moves. That is the curse.
- **A neuron is a ridge** $\sigma(w^\top x + b)$: a soft step aimed along a *learned* direction w . The net spends its M units where f varies — so error tracks f 's *intrinsic* complexity, not the dimension N .



For economists: this is why deep learning cracks high-dimensional dynamic models — heterogeneous-agent macro, many assets and shocks — exactly where Chebyshev and tensor grids break down (Lec. 4–6; the DeepRamsey case).

DNN = Linear Transformation + Nonlinear Activation

- **Linear Transformation:**

- Each neuron performs a weighted sum of inputs: projects data onto a new axis
- The set of transformations across a layer creates a *new representation*
- Enables: flexible feature representation, information compression, interaction capture

- **Nonlinear Activation:**

- Allows the network to capture nonlinearity and complex patterns
- Creates curved decision boundaries in the feature space
- **Essential** for the universal approximation property

- **Together:** Each layer applies an affine map $Wx + b$ then a pointwise nonlinearity $\sigma(\cdot)$. Stacking layers composes these maps:

$$f(x) = \sigma_L \circ A_L \circ \cdots \circ \sigma_1 \circ A_1(x)$$

where $A_\ell(x) = W_\ell x + b_\ell$.

- **Next (T3):** how to *train* this composition — forward pass, backpropagation, gradient descent.